# Sri Lanka Institute of Information Technology

**IT1040 - Fundamentals of Computing**

**Year 1, Semester 1- 2024**

**<Water Tank Monitoring System>**

Progress Report

<P10-5>

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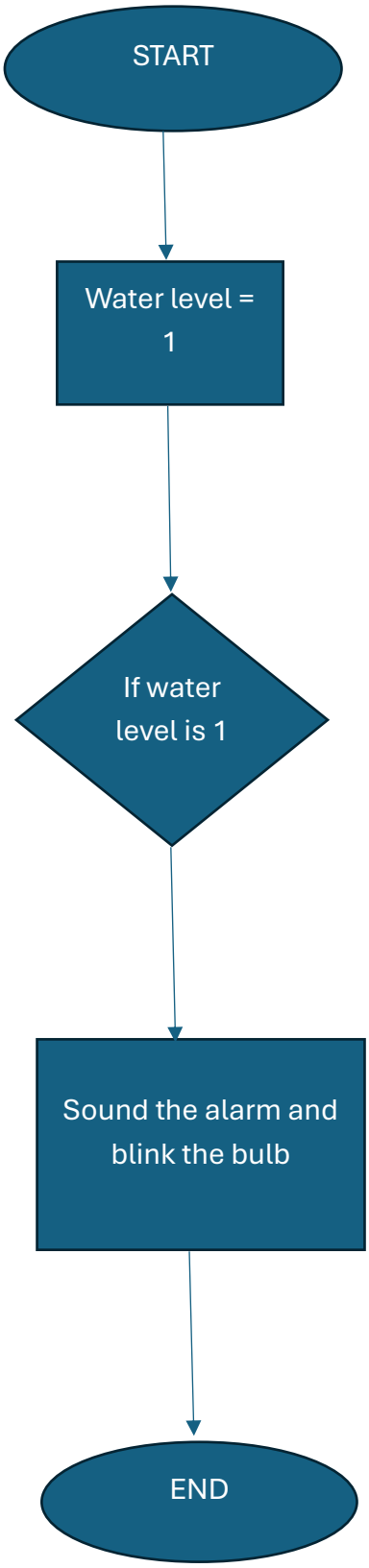
AGENDA

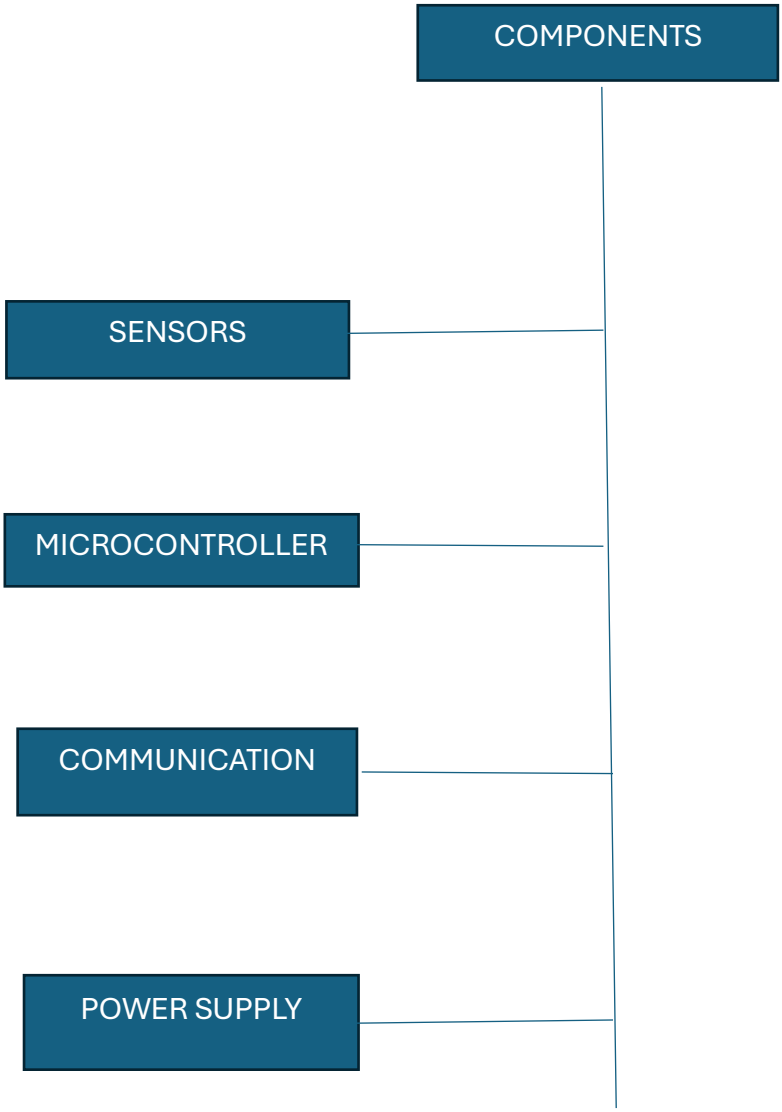
- INTRODUCTION
- SYSTEM ARCHITECTURE
- WORK BREAKDOWN STRUCTURE
- PROOF OF WORK

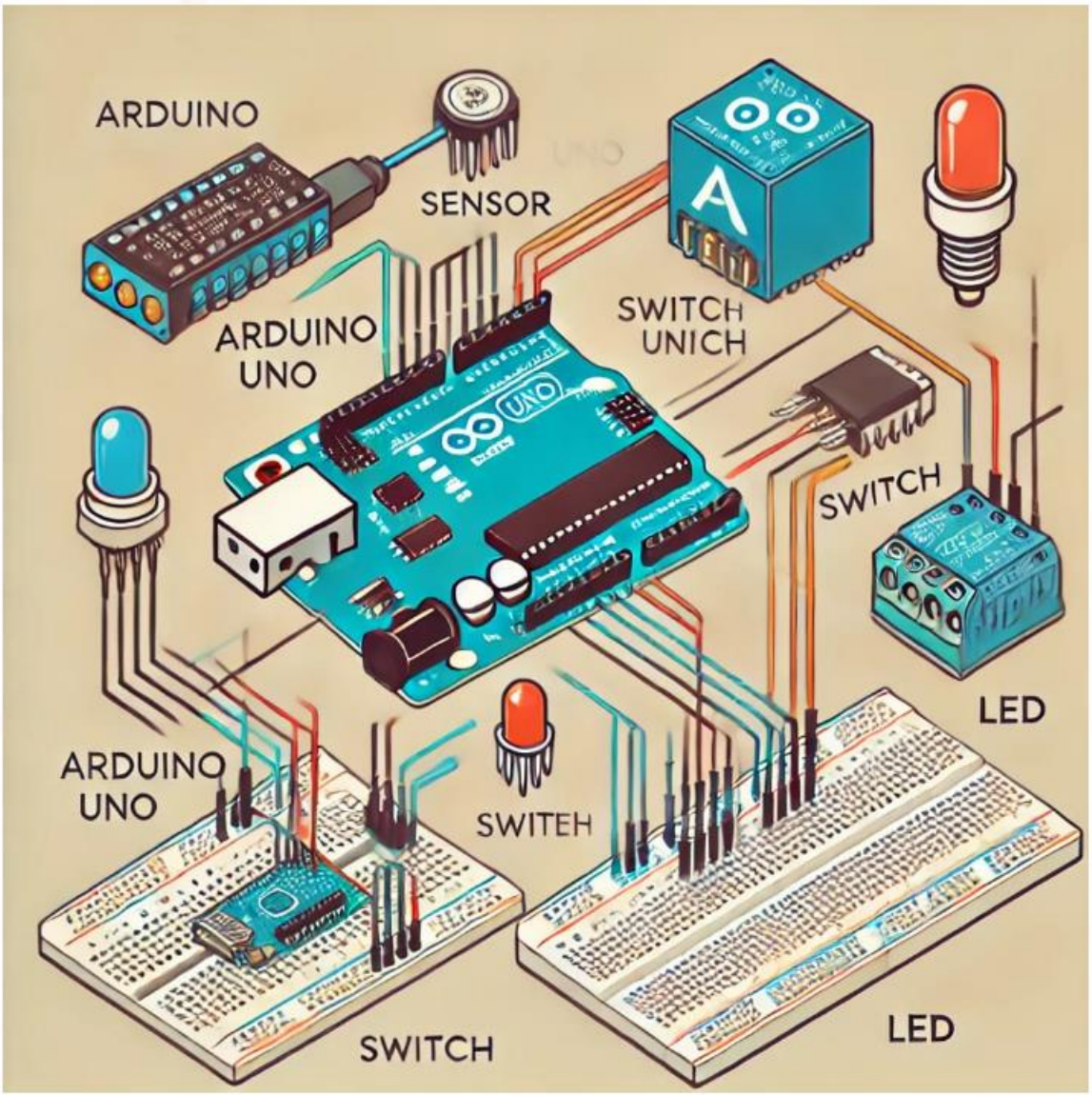
# 1.INTRODUCTION

Developing a water tank monitoring system based on an IOT system is a quiet ctitical and a challenging task. In our project, we mainly focus on how to measure the water level of the water tank. Most importantly to reduce the unnecessary wastage of  water from the water tank when the water level of the water tank increases. And also with this developing system provides real-time updates of the water level in the water tank. With the use of Arduino microcontrollers and sensors the solution ensures more efficient and reliable water management.

# 2.SYSTEM ARCHITECTURE




COMPONENTS
SENSORS
MICROCONTROLLER
COMMUNICATION
POWER SUPPLY




The system will have both hardware and software elements, with communication between them.

System Architecture Diagram Overview

Components:

1. Arduino Uno Board – Central controller.

2. Sensor – Input component (e.g., temperature sensor, motion sensor).

3. Switch – Manual input component to control certain operations.

4. Relay – Controls higher voltage devices.

5. LED – Output indicator (shows status of relay or system).

6. Breadboard – Used for circuit connections.

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Hardware Level

1. Sensor → Arduino Uno

The sensor continuously sends data to the Arduino board. The sensor can be anything like a temperature sensor, motion sensor, etc.

2. Switch → Arduino Uno

A switch is connected to the input pin on the Arduino and allows manual toggling of the system.

3. Arduino Uno → Relay

The Arduino controls the relay based on sensor input or switch state. The relay is used to operate higher power loads or external devices.

4. Arduino Uno → LED

The Arduino sends a signal to the LED to turn it on or off as an indicator, typically reflecting the status of the relay or other conditions.

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Software Flow

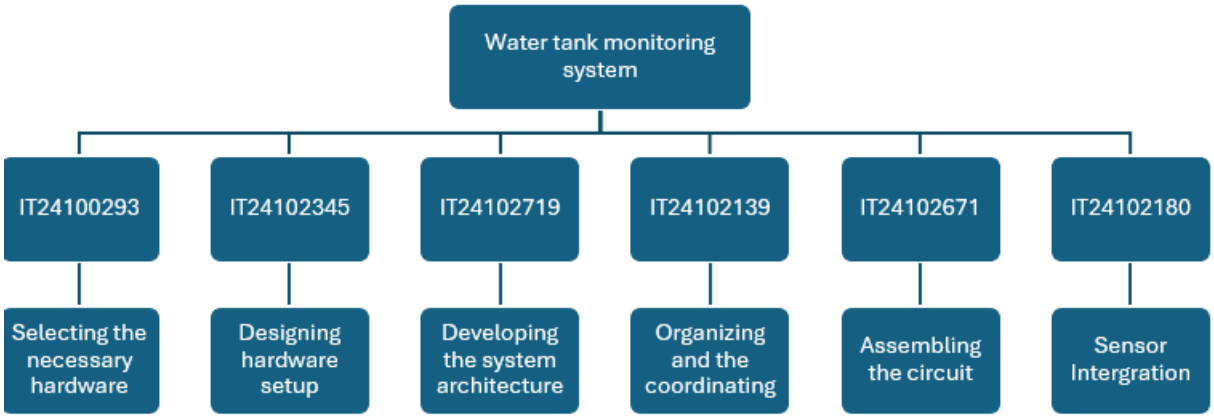
Sensor Data: Arduino processes the data received from the sensor.

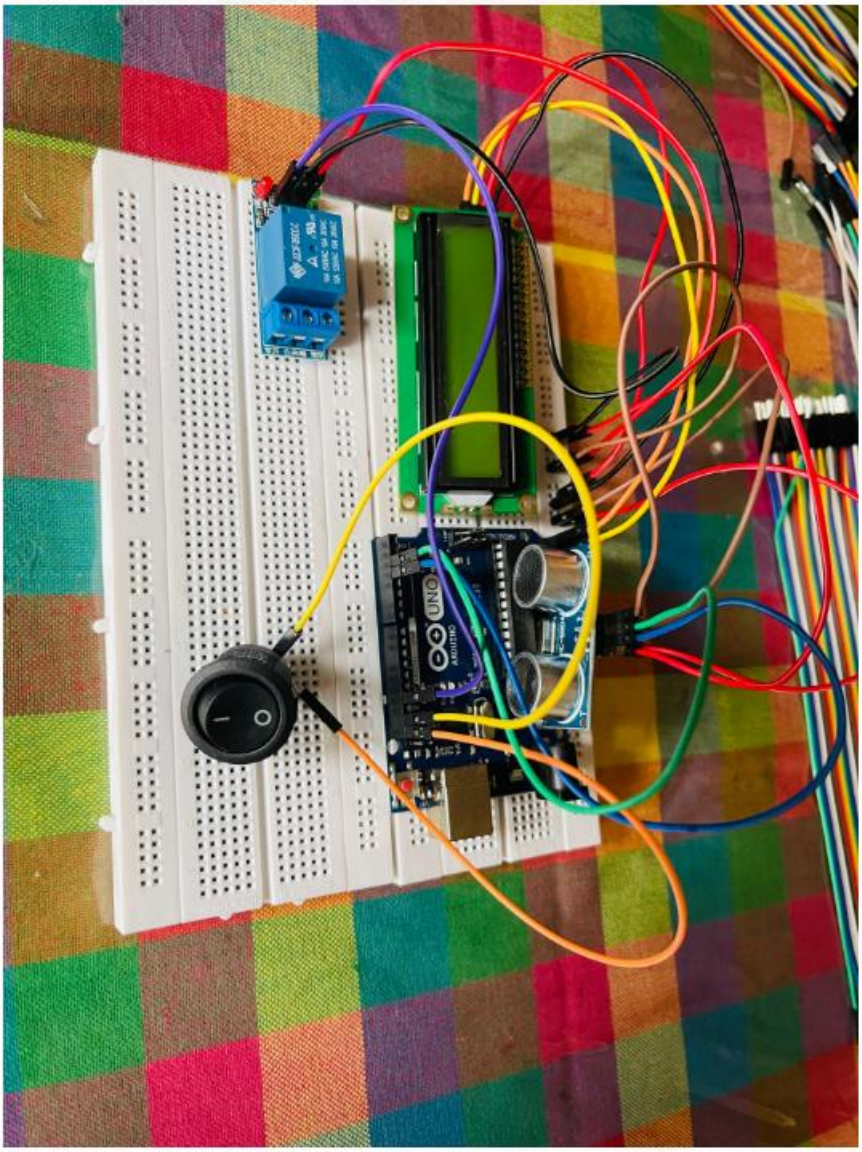
Decision Making: Based on the sensor data and switch state, the Arduino decides whether to activate the relay.

Relay Control: If conditions are met, the Arduino sends a signal to trigger the relay.

LED Feedback: The LED provides visual feedback based on the relay's state or other system condition

# 3.Work Breakdown Structure




Our teammates actively participate in this project to provide a more efficient water level monitoring system.

# 4.Proof of work